KATIE CARLSON

Location: San Luis Obispo County, CA | **Phone:** (805) 715-5728 | **Email:** kcarls24@calpoly.edu
Citizenship: United States Citizen | **Clearance/Status:** Eligible | **Veterans' Preference:** N/A
GitHub Portfolio: GitHub Credentials | **LinkedIn:** KatieCarlson

OBJECTIVE

Detail-driven Computer Science, Physics, and Aerospace Engineering scholar with active NASA-funded leadership credentials as a Command and Data Handling (C&DH) Lead. Seeking a Summer 2027 NASA OSTEM Internship within the Mission Operations Directorate (MOD), Spacecraft Software, or Flight Operations Divisions. Intent on applying real-time telemetry systems engineering, deterministic data routing, and active command architectures toward future Flight Director and Astronaut candidate selection tracks.

EDUCATION & ACADEMIC ACCREDITATION

CALIFORNIA POLYTECHNIC STATE UNIVERSITY (CAL POLY), SAN LUIS OBISPO

Blended B.S. / M.S. in Computer Science (Incoming Transfer Fall 2028 / Blended Track)

- **Minors:** Physics and Aerospace Engineering | **Expected Graduation:** Spring 2032
- **Focus Areas:** Embedded Systems, Real-Time Spacecraft Software Architecture, Orbital Mechanics, Computational Astrophysics.

CUESTA COLLEGE, SAN LUIS OBISPO, CA

Undergraduate Coursework in Computer Science & Physics

- **Relevant Coursework:** Data Structures & Algorithms, Assembly Language Architecture, Classical Mechanics, Physics for Scientists & Engineers (Mechanics, Electricity & Magnetism, Modern Physics), Linear Algebra, Differential Equations, Multivariable Calculus.

AEROSPACE WORKFORCE & SPACE HARDWARE EXPERIENCE

NASA L'SPACE MISSION CONCEPT ACADEMY (MCA)

Command and Data Handling (C&DH) Engineer Lead | Team 15 "Astra Continuum" | *Fall 2026 – Present*

- Spearheaded the architectural design of the Command and Data Handling (C&DH) sub-system for a NASA-templated planetary science mission concept, adhering to NASA Procedural Requirements (NPR) and telemetry standards.

- Engineered embedded telemetry routing algorithms and simulated real-time data handling loops to manage bandwidth constraints, onboard storage buffers, and downlinking requirements from scientific payloads to communication transceivers.
- Moderated multi-disciplinary technical trade studies evaluating microcontrollers, radiation-hardened FPGA architectures, power budgets, mass margins, and flight software operating system constraints (FreeRTOS / cFS).
- Synthesized complex system requirements and managed technical risk matrices (Failure Modes and Effects Analysis - FMEA) across structural, power, thermal, and payload interfaces to guarantee mission success.
- Drafted and finalized mission-critical C&DH technical documentation, presenting systems engineering trade-offs during structured Preliminary Design Reviews (PDR) and Critical Design Reviews (CDR).

POLYSAT & SPACECRAFT HARDWARE PREPARATION

C&DH & Flight Software Development Track | *San Luis Obispo, CA*

- Formulated telemetry processing schemas and data logging architectures optimized for CubeSat flight computers and static web simulation testbeds.
- Specialized in Inertial Measurement Unit (IMU) data assimilation, dead reckoning software models, and closed-loop Attitude Determination and Control Systems (ADCS) simulation parameters.
- Collaborated with assembly, integration, and testing (AIT) workflows to understand physical environmental testing constraints including vibration, thermal vacuum (TVAC), and power cycling protocols.

SOFTWARE ARCHITECTURE & RESEARCH PROJECTS

SYSTEM ARCHITECT & CREATOR | ETERNAL COSMIC INFINITY PORTAL

Open-Source Aerospace & Computational Physics Ecosystem | *2026*

- Engineered and maintained a suite of high-performance web applications and simulators modeling spacecraft trajectories, orbital parameters, and computational linguistics using HTML5, Tailwind CSS, and Native JavaScript.
- **Live Orbital Simulators (Astra Luminous / Astra Trackers):** Modeled Low Earth Orbit (LEO) velocity trajectories (~7.67 km/s), suborbital ballistic arcs, and coplanar Hohmann Transfer maneuvers to GEO, providing real-time vector visualization.
- **Siemens NX 1984 Windows 11 ARM Optimizer:** Architected an automated PowerShell deployment script resolving storage exhaustion and virtual GPU rendering locks for Siemens NX CAD execution on Apple Silicon (M1–M4) hardware running virtualized Windows 11 ARM via Microsoft Prism translation layer.
- **1,000-Ton Deep-Space Crewed Cruiser CAD:** Designed a parametric spacecraft assembly featuring a 200MWe nuclear thermal rejection plant, 40-grid ion thruster matrix ($I_{sp} = 3,000s$), sacrificial Whipple shielding, and deployable thermal radiators.

- **Pāṇinian Rule JSON Transducer:** Programmed a deterministic linguistic validation engine translating ancient Sanskrit grammatical formulas into machine-readable JSON Schema Draft-07 validation payloads for natural language processing (NLP) research.

GUEST LECTURING & SCIENTIFIC OUTREACH

ASTRO-NOT ASTRO-TALKS WEBINAR SERIES

Keynote Speaker & Guest Lecturer: "Rocket Science & Orbital Mechanics" | *November 2026*

- Delivered an international technical webinar to 80+ undergraduate students and researchers on rocket propulsion, gravity turns, Tsiolkovsky rocket equation dynamics, and the physics of entering orbit versus suborbital trajectories.
- Demonstrated live interactive orbital simulations showing orbital circularization burns at apogee, perigee adjustments, and relative motion phasing maneuvers.

TECHNICAL SKILLS MATRIX

Category	Technical Capabilities & Tools
Programming Languages	C++, Python, C, Assembly (x86/ARM), JavaScript (ES6+), HTML5, CSS3, PowerShell
Systems & Software	Git, GitHub Actions, Linux/Unix, JSON Schema, Siemens NX 1984 CAD, JMARS (Planetary GIS), Parallels / ARM64 Virtualization
Aerospace Methodologies	C&DH Systems Engineering, PDR/CDR Execution, Failure Modes and Effects Analysis (FMEA), Telemetry Loop Design, Trade Studies, Flight Software Architecture, Subsystem Interfacing
Physics & Math	Orbital Mechanics (Keplerian Elements, Δv Budgeting, Hohmann Transfers), Classical Mechanics, Linear Algebra, Differential Equations, Wave Physics & Acoustic Resonance

CREDENTIALS & CERTIFICATIONS

- **NASA L'SPACE Mission Concept Academy Certificate of Completion** – NASA / (Dec 2026)
- **200-Hour Certified Yoga Instructor & Sound Healing Practitioner** – Accredited Vinyasa/Hatha & Acoustic Resonance Specialist (Dec 2026)
- **AIIS Language Program & CAORC Grant Applicant** – Classical Language & Computational Linguistics Research in India (Spring 2027 Cohort)